NEET UG CHEMISTRY TUTORIAL

Classification of Elements and Periodicity in Properties

Syllabus Context: Inorganic Chemistry Foundation — Historical Development of Periodic Table, Modern Periodic Law, IUPAC Nomenclature ($Z > 100$), Electronic Configuration (s, p, d, f Blocks), and Periodic Trends in Physical and Chemical Properties.

1. Learning Objectives and Chapter Overview

After studying this tutorial, you will be able to:

- Appreciate how the concept of grouping elements according to their properties led to the development of the Periodic Table.
- Understand the Mendeleev Periodic Law and the Modern Periodic Law.
- Understand the significance of atomic number and electronic configuration as the basis for periodic classification.
- Name the elements with atomic number $Z > 100$ according to systematic IUPAC nomenclature.
- Classify elements into s-, p-, d-, and f-blocks and learn their main characteristics.
- Recognise and explain the periodic trends in physical and chemical properties of elements.
- Compare the reactivity of elements and correlate it with their occurrence in nature.
- Explain the relationship between ionization enthalpy and metallic character.
- Use scientific vocabulary appropriately to communicate ideas related to atomic/ionic radii, ionization enthalpy, electron gain enthalpy, electronegativity, and valence of elements.

> *"The Periodic Table is arguably the most important concept in chemistry, both in principle and in practice. It is the everyday support for students, it suggests new avenues of research to professionals, and it provides a succinct organization of the whole of chemistry. It is a remarkable demonstration of the fact that the chemical elements are not a random cluster of entities but instead display trends and lie together in families. An awareness of the Periodic Table is essential to anyone who wishes to disentangle the world and see how it is built up from the fundamental building blocks of the chemistry, the chemical elements."*
> ***— Glenn T. Seaborg***

Why Do We Need to Classify Elements?

Elements are the basic units of all types of matter. Historically, the discovery of elements progressed rapidly:

- In **1800**, only **31 elements** were known.
- By **1865**, the number of identified elements had more than doubled to **63**.
- At present, **114 to 118 elements** are known (with recent elements being man-made through nuclear synthesis).

With such a large number of elements, it is very difficult to study individually the chemistry of all these elements and their innumerable compounds. To ease this problem, scientists searched for a systematic way to organise their knowledge by classifying the elements. Classification not only rationalizes known chemical facts about elements, but also predicts new ones for undertaking further study.

2. Genesis of Periodic Classification

2.1 Dobereiner's Triads (1829)

The German chemist **Johann Dobereiner** in the early 1800s was the first to consider the idea of trends among properties of elements. By 1829, he noted a similarity among the physical and chemical properties of several groups of three elements called **Triads**.

Law of Triads: In each triad, the middle element had an atomic weight about half-way between the atomic weights of the other two, and the properties of the middle element were in between those of the other two members. Because this relationship seemed to work only for a few elements, it was dismissed as a coincidence.

Dobereiner's Triads

Triad 1 (Alkali Metals)	Atomic Weight	Triad 2 (Alkaline Earth Metals)	Atomic Weight	Triad 3 (Halogens)	Atomic Weight
Li	7	Ca	40	Cl	35.5
Na	23	Sr	88	Br	80
K	39	Ba	137	I	127

2.2 De Chancourtois' Cylindrical Table (1862)

A French geologist, **A.E.B. de Chancourtois**, arranged the then-known elements in order of increasing atomic weights and made a cylindrical table of elements to display the periodic recurrence of properties. This did not attract much attention.

2.3 Newlands' Law of Octaves (1865)

The English chemist **John Alexander Newlands** propounded the **Law of Octaves**. He arranged the elements in increasing order of their atomic weights and noted that every eighth element had properties similar to the first element (just like every eighth note resembles the first in musical octaves).

- **Limitation:** Newlands' Law of Octaves seemed to be true only for elements up to **calcium (Ca)**.
- **Recognition:** Although not widely accepted at that time, he was later awarded the **Davy Medal in 1887** by the Royal Society, London.

Newlands' Octaves

Row	1	2	3	4	5	6	7
Element At. wt.	Li 7	Be 9	B 11	C 12	N 14	O 16	F 19
Element At. wt.	Na 23	Mg 24	Al 27	Si 29	P 31	S 32	Cl 35.5
Element At. wt.	K 39	Ca 40	–	–	–	–	–

2.4 Lothar Meyer and Dmitri Mendeleev (1869)

Working independently in 1869, Russian chemist **Dmitri Mendeleev** (1834–1907) and German chemist **Lothar Meyer** (1830–1895) proposed that on arranging elements in increasing order of their atomic weights, similarities appear in physical and chemical properties at regular intervals.

- **Lothar Meyer's Contribution:** He plotted physical properties such as *atomic volume, melting point, and boiling point against atomic weight* and obtained a periodically repeated pattern with changing lengths. By 1868, he had developed a table closely resembling the Modern Periodic Table, but his work was published after Mendeleev's.
- **Mendeleev's Periodic Law:** Published for the first time by Mendeleev, it states:

> **Mendeleev's Periodic Law:** "The properties of the elements are a periodic function of their atomic weights."

Key Features of Mendeleev's Periodic Table:

1. **Groups and Series:** Elements were arranged in horizontal rows (called *series*) and vertical columns (called *groups*) in order of increasing atomic weights so that elements with similar properties occupied the same vertical group.
2. **Empirical Formulas as Criteria:** Mendeleev relied heavily on similarities in the empirical formulas and properties of compounds formed by the elements (such as oxides $\mathrm{R_2O}, \mathrm{RO}, \mathrm{R_2O_3}, \mathrm{RO_2}, \mathrm{R_2O_5}, \mathrm{RO_3}, \mathrm{R_2O_7}, \mathrm{RO_4}$ and hydrides $\mathrm{RH_4}, \mathrm{RH_3}, \mathrm{RH_2}, \mathrm{RH}$).
3. **Priority to Similarity in Properties over Atomic Weight Order:** Where strict atomic weight order did not fit chemical properties, he ignored the order of atomic weights (assuming atomic weight measurements might be incorrect). For example, **Iodine (I, atomic weight 126.9)** has a lower atomic weight than **Tellurium (Te, atomic weight 127.6 in Group VI)**, yet Mendeleev placed Iodine in Group VII along with $\mathrm{F}, \mathrm{Cl}, \mathrm{Br}$ because of similarities in chemical properties.
4. **Prediction of Undiscovered Elements:** To keep elements with similar properties in the same group, Mendeleev left several gaps for undiscovered elements. He left a gap under aluminium (Al) and called it **Eka-Aluminium** (later discovered as **Gallium,** Ga) and a gap under silicon (Si) called **Eka-Silicon** (later discovered as **Germanium,** Ge), quantitatively predicting their physical and chemical properties with remarkable accuracy.

Mendeleev's Predictions vs. Experimentally Found Properties for Eka-Aluminium and Eka-Silicon

Property	Eka-aluminium (Predicted)	Gallium (Ga, Found)	Eka-silicon (Predicted)	Germanium (Ge, Found)
Atomic weight	68	70	72	72.6
Density / ($\mathrm{g\,cm^{-3}}$)	5.9	5.94	5.5	5.36
Melting point / K	Low	302.93	High	1231
Formula of oxide	$\mathrm{E_2O_3}$	$\mathrm{Ga_2O_3}$	$\mathrm{EO_2}$	$\mathrm{GeO_2}$
Formula of chloride	$\mathrm{ECl_3}$	$\mathrm{GaCl_3}$	$\mathrm{ECl_4}$	$\mathrm{GeCl_4}$

3. Modern Periodic Law and the Present Form of the Periodic Table

In 1913, the English physicist **Henry Moseley** observed regularities in the characteristic *X-ray spectra* of the elements. A plot of $\sqrt{\nu}$ (where ν is the frequency of X-rays emitted) against **atomic number (Z)** gave a straight line, whereas the plot of $\sqrt{\nu}$ versus atomic mass did not.

Moseley thereby demonstrated that **atomic number (Z) is a more fundamental property of an element than its atomic mass**. Mendeleev's Periodic Law was accordingly modified into the **Modern Periodic Law**:

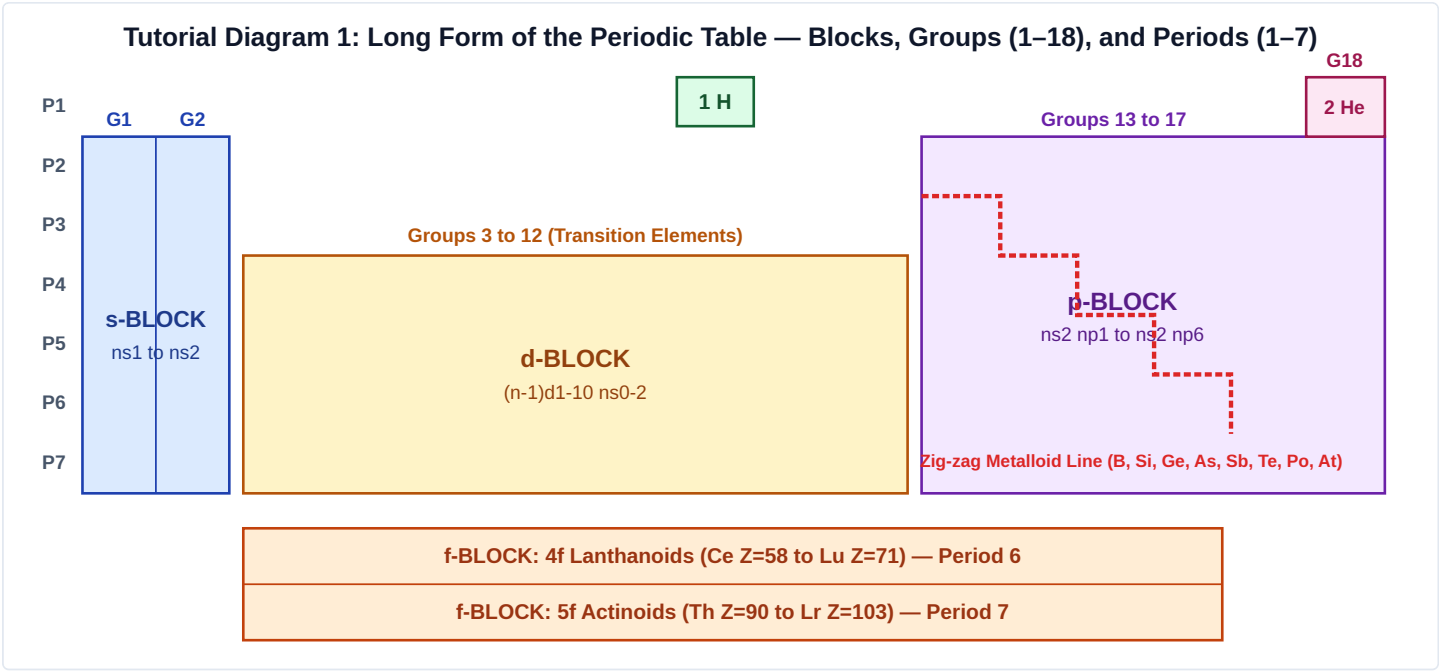
> **Modern Periodic Law:** "The physical and chemical properties of the elements are periodic functions of their atomic numbers."

The atomic number (Z) is equal to the nuclear charge (number of protons) or the number of electrons in a neutral atom. Thus, the Periodic Law is essentially the consequence of the **periodic variation in electronic configurations**, which determine the physical and chemical properties of elements and their compounds.

Note on Naturally Occurring Elements: There are 94 naturally occurring elements (neptunium, $Z = 93$, and plutonium, $Z = 94$, like actinium and protactinium, are also found in pitchblende—an ore of uranium).

3.1 Structure of the "Long Form" of the Periodic Table

- **Periods (Horizontal Rows):** There are altogether **7 periods**. The period number corresponds to the highest principal quantum number (n) of the elements in that period.
  - 1st Period ($n = 1$): **2 elements** (H to He)
  - 2nd Period ($n = 2$): **8 elements** (Li to Ne)
  - 3rd Period ($n = 3$): **8 elements** (Na to Ar)
  - 4th Period ($n = 4$): **18 elements** (K to Kr)
  - 5th Period ($n = 5$): **18 elements** (Rb to Xe)
  - 6th Period ($n = 6$): **32 elements** (Cs to Rn, including 14 lanthanoids)
  - 7th Period ($n = 7$): Theoretical maximum of **32 elements** (up to $Z = 118$, including 14 actinoids)
- **Groups (Vertical Columns):** Elements having similar outer electronic configurations in their atoms are arranged in vertical columns called **groups** or **families**. According to the 1984 IUPAC recommendation, groups are numbered from **1 to 18**, replacing the older notation of $\mathrm{IA} \ldots \mathrm{VIIA}, \mathrm{VIII}, \mathrm{IB} \ldots \mathrm{VIIB}$ and 0.
- **Separate Bottom Panels:** 14 elements of the sixth period (**Lanthanoids**, Ce $Z = 58$ to Lu $Z = 71$) and 14 elements of the seventh period (**Actinoids**, Th $Z = 90$ to Lr $Z = 103$) are placed in separate panels at the bottom to maintain the table's structure and keep elements with similar properties in a single column. Glenn T. Seaborg's mid-20th-century work on plutonium (1940) and transuranium elements ($Z = 94$ to 102) led to this reconfiguration, earning him the 1951 Nobel Prize in Chemistry; element 106 is named **Seaborgium (Sg)** in his honour.



Division of the Periodic Table into s-, p-, d-, and f-blocks based on the orbital receiving the last electron, along with the zig-zag metalloid border in the p-block.

4. IUPAC Nomenclature of Elements with Atomic Numbers $Z > 100$

New elements with very high atomic numbers are extremely unstable and synthesized in minute quantities (sometimes only a few atoms), leading to historical naming controversies. For example, both American and Soviet scientists claimed credit for discovering **element 104**: the Americans named it **Rutherfordium** whereas the Soviets named it **Kurchatovium**.

To avoid such problems, IUPAC recommended that until a new element's discovery is proved and its official name is recognized, a **systematic three-letter symbol and temporary name** be derived directly from the digits of its atomic number:

1. Combine the numerical roots for each digit of the atomic number in order.
2. Add the suffix **"-ium"** at the end (if "enn" precedes "nil", drop one 'n'; if "bi" or "tri" precedes "ium", drop one 'i').
3. Capitalize the first letter of the three-letter symbol.

Numerical Roots for IUPAC Systematic Nomenclature of Elements

Digit	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
Root Name	nil	un	bi	tri	quad	pent	hex	sept	oct	enn
Abbreviation	n	u	b	t	q	p	h	s	o	e

Systematic and Official IUPAC Names of Elements with $Z > 100$

Atomic Number (Z)	Systematic IUPAC Name	Systematic Symbol	Official IUPAC Name	Official Symbol
101	Unnilunium	Unu	Mendelevium	Md
102	Unnilbium	Unb	Nobelium	No
103	Unniltrium	Unt	Lawrencium	Lr
104	Unnilquadium	Unq	Rutherfordium	Rf
105	Unnilpentium	Unp	Dubnium	Db
106	Unnilhexium	Unh	Seaborgium	Sg
107	Unnilseptium	Uns	Bohrium	Bh
108	Unniloctium	Uno	Hassium	Hs
109	Unnilennium	Une	Meitnerium	Mt
110	Ununnillium	Uun	Darmstadtium	Ds
111	Unununnium	Uuu	Rontgenium	Rg
112	Ununbium	Uub	Copernicium	Cn
113	Ununtrium	Uut	Nihonium	Nh
114	Ununquadium	Uuq	Flerovium	Fl
115	Ununpentium	Uup	Moscovium	Mc
116	Ununhexium	Uuh	Livermorium	Lv
117	Ununseptium	Uus	Tennessine	Ts
118	Ununoctium	Uuo	Oganesson	Og

5. Electronic Configurations and Types of Elements (s, p, d, f Blocks)

5.1 Electronic Configurations in Periods

The period number indicates the value of n for the outermost or valence shell. The number of elements in each period is **twice the number of atomic orbitals available** in the energy level that is being filled:

- **1st Period ($n = 1$):** Fills $1s$ orbital (1 orbital $\Rightarrow$ 2 elements): $\mathrm{H}\,(1s^1)$ and $\mathrm{He}\,(1s^2)$, completing the K shell.
- **2nd Period ($n = 2$):** Fills $2s$ and $2p$ orbitals (4 orbitals $\Rightarrow$ 8 elements): starts at $\mathrm{Li}\,(1s^2 2s^1)$, $\mathrm{Be}\,(1s^2 2s^2)$, B to $\mathrm{Ne}\,(2s^2 2p^6)$, completing the L shell.
- **3rd Period ($n = 3$):** Fills $3s$ and $3p$ orbitals (4 orbitals $\Rightarrow$ 8 elements): from Na to Ar.
- **4th Period ($n = 4$):** Fills $4s$, $3d$, and $4p$ orbitals (9 orbitals $\Rightarrow$ 18 elements): starts at $\mathrm{K}\,(4s^1)$ and $\mathrm{Ca}\,(4s^2)$, followed by the $3d$ **transition series** from $\mathrm{Sc}\,(Z = 21, 3d^1 4s^2)$ to $\mathrm{Zn}\,(Z = 30, 3d^{10} 4s^2)$, and ends at Kr with the filling of $4p$ orbitals.
- **5th Period ($n = 5$):** Fills $5s$, $4d$, and $5p$ orbitals (9 orbitals $\Rightarrow$ 18 elements): starts at Rb, contains the $4d$ **transition series** starting at $\mathrm{Y}\,(Z = 39)$, and ends at Xe with the filling of $5p$ orbitals.
- **6th Period ($n = 6$):** Fills $6s$, $4f$, $5d$, and $6p$ orbitals (16 orbitals $\Rightarrow$ 32 elements): filling of $4f$ orbitals begins at **Cerium (Ce, $Z = 58$)** and ends at **Lutetium (Lu, $Z = 71$)** to give the $4f$-inner transition series (**Lanthanoid series**).
- **7th Period ($n = 7$):** Fills $7s$, $5f$, $6d$, and $7p$ orbitals (16 orbitals $\Rightarrow$ 32 elements): ends at $Z = 118$ (noble gas family). Filling of $5f$ orbitals after **Actinium (Ac, $Z = 89$)** from $\mathrm{Th}\,(Z = 90)$ to $\mathrm{Lr}\,(Z = 103)$ gives the $5f$-inner transition series (**Actinoid series**), which includes most man-made radioactive elements.

5.2 Groupwise Electronic Configurations

Elements in the same vertical column (group) have similar valence shell electronic configurations, the same number of electrons in the outer orbitals, and therefore similar chemical properties. For example, all Group 1 elements (alkali metals) have an ns^1 valence shell configuration:

- $\mathrm{Li}\,(Z = 3) : 1s^2 2s^1$ or $[\mathrm{He}]2s^1$
- $\mathrm{Na}\,(Z = 11) : 1s^2 2s^2 2p^6 3s^1$ or $[\mathrm{Ne}]3s^1$
- $\mathrm{K}\,(Z = 19) : 1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$ or $[\mathrm{Ar}]4s^1$
- $\mathrm{Rb}\,(Z = 37) : 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 5s^1$ or $[\mathrm{Kr}]5s^1$
- $\mathrm{Cs}\,(Z = 55) : 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^2 5p^6 6s^1$ or $[\mathrm{Xe}]6s^1$
- $\mathrm{Fr}\,(Z = 87) : [\mathrm{Rn}]7s^1$

5.3 Classification into s-, p-, d-, and f-Blocks

> **Two Important Exceptions in Block Placement:**
> 1. **Helium (He, $1s^2$):** Strictly belongs to the s-block, but is placed in the p-block (Group 18) because it has a completely filled valence shell ($1s^2$) and exhibits properties characteristic of other noble gases.
> 2. **Hydrogen (H, $1s^1$):** Has one s-electron and can lose it like Group 1 alkali metals, or gain one electron to achieve a noble gas arrangement like Group 17 halogens. Hence, it is placed separately at the top of the Periodic Table.

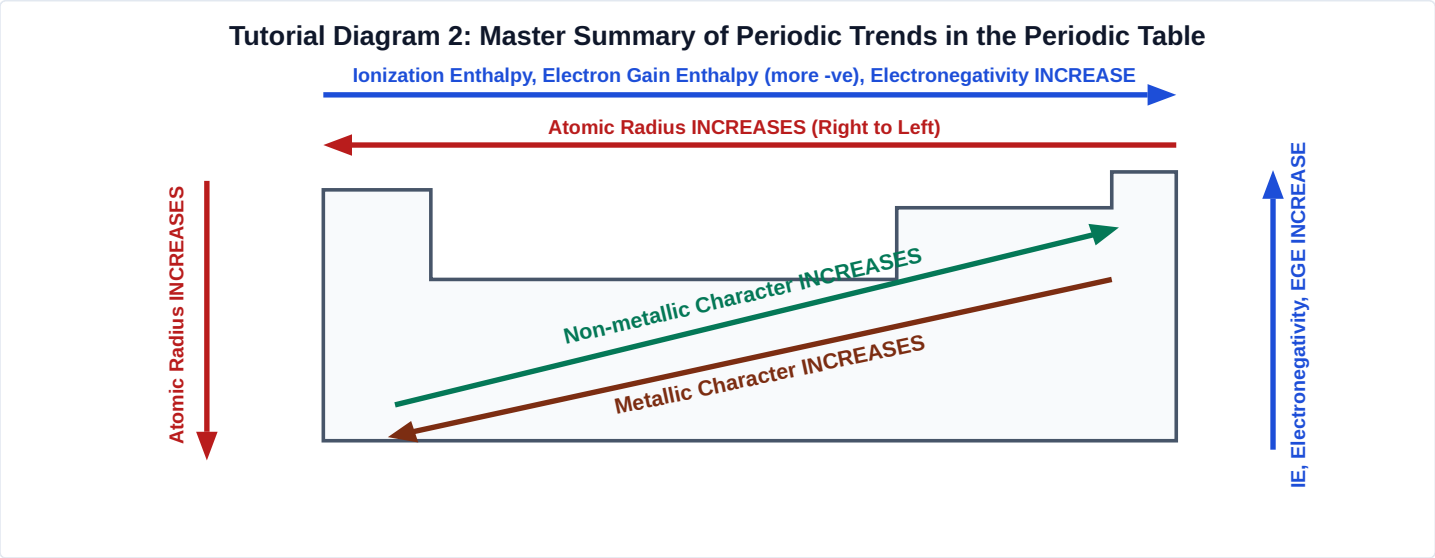
Characteristics of s-, p-, d-, and f-Block Elements

Block	Groups	General Outer Configuration	Key Characteristics and Exceptions
s-Block	Groups 1 & 2	ns^{1-2}	• Group 1: Alkali metals (ns^1), form $+1$ ions. • Group 2: Alkaline earth metals (ns^2), form $+2$ ions. • Reactive metals with low ionization enthalpies; never found pure in nature. • Metallic character and reactivity increase down the group. • Compounds are predominantly ionic, **except Li and Be** (which show pronounced covalent character).
p-Block	Groups 13 to 18	ns^2np^{1-6}	• Together with s-block elements, called **Representative Elements** or **Main Group Elements**. • Group 18: **Noble gases** (ns^2np^6, except $\mathrm{He}\ 1s^2$), closed valence shell, very low chemical reactivity. • Group 17: **Halogens**; Group 16: **Chalcogens**. Both have highly negative electron gain enthalpies and readily gain 1 or 2 electrons. • Includes metals, non-metals, and metalloids.
d-Block (Transition Elements)	Groups 3 to 12	$(n-1)d^{1-10}ns^{0-2}$ (Note: Pd is $4d^{10}5s^0$)	• Characterized by filling of inner d orbitals. • All are metals; mostly form coloured ions, exhibit variable valence (oxidation states), paramagnetism, and act as catalysts. • Form a bridge between active s-block metals and less active Group 13/14 elements. • **Exception:** Zn, Cd, and Hg have $(n-1)d^{10}ns^2$ configuration and do *not* show most properties of transition elements.
f-Block (Inner-Transition Elements)	Between Group 3 & 4 (Periods 6 & 7)	$(n-2)f^{1-14}(n-1)d^{0-1}ns^2$	• **Lanthanoids:** Ce $(Z=58)$ to Lu $(Z=71)$. • **Actinoids:** Th $(Z=90)$ to Lr $(Z=103)$. • All are metals with similar properties within each series. • Early actinoid chemistry is more complicated than lanthanoids due to a large number of possible oxidation states. • Actinoids are radioactive. Elements after Uranium ($Z>92$) are called **Transuranium Elements**.

5.4 Metals, Non-Metals, and Metalloids

- **Metals (>78% of all known elements):** Located on the left side of the Periodic Table. Usually solids at room temperature (*exceptions:* **mercury (Hg)** is a liquid; **gallium (Ga, m.p. $303\ \mathrm{K}$)** and **caesium (Cs, m.p. $302\ \mathrm{K}$)** have very low melting points). Have high melting and boiling points, are good conductors of heat and electricity, and are malleable and ductile.
- **Non-Metals (<20 elements):** Located at the top right-hand side of the Periodic Table. Usually solids or gases at room temperature (bromine is a liquid) with low melting and boiling points (*exceptions:* **boron (B)** and **carbon (C)** have high melting/boiling points). Poor conductors of heat and electricity; brittle solids (neither malleable nor ductile).
- **Metalloids (Semi-metals):** Elements bordering the thick zig-zag line running diagonally across the p-block (e.g., **silicon (Si), germanium (Ge), arsenic (As), antimony (Sb), and tellurium (Te)**) show properties characteristic of both metals and non-metals.

6. Periodic Trends in Physical Properties



Summary of periodic variations in atomic radius, ionization enthalpy (IE), electron gain enthalpy (EGE), electronegativity, and metallic/non-metallic character.

6.1 Atomic Radius

Because an atom is extremely small (~ 1.2 Å, i.e., 1.2×10^{-10} m in radius) and its surrounding electron cloud does not have a sharp boundary, the size of an individual isolated atom cannot be measured directly. Instead, it is estimated from internuclear distances in combined states using X-ray or spectroscopic methods:

- **Covalent Radius (for non-metals):** Half the internuclear distance between two identical atoms bound together by a single covalent bond in a molecule.
  Example: The bond distance in the chlorine molecule (Cl_2) is 198 pm, so the atomic (covalent) radius of chlorine is $198/2 = 99$ pm.
- **Metallic Radius (for metals):** Half the internuclear distance separating the metal cores in a metallic crystal.
  Example: The distance between two adjacent copper atoms in solid copper is 256 pm, so the metallic radius of copper is $256/2 = 128$ pm.
- **Van der Waals Radius (for Noble Gases):** Because noble gases are monoatomic, their non-bonded radii values are very large and should be compared only with the *van der Waals radii* of other elements, not covalent radii.

Periodic Trends in Atomic Radius:

1. **Across a Period (Left to Right): Decreases.** Within a period, outer electrons are in the same valence shell while the nuclear charge (Z) increases. Consequently, the **effective nuclear charge (Z_{eff}) increases**, drawing the electron cloud closer to the nucleus.
2. **Down a Group (Top to Bottom): Increases.** As we descend a group, the principal quantum number (n) increases and inner energy levels filled with core electrons **shield (screen)** the outer valence electrons from the pull of the nucleus.

Atomic Radii (pm) Across Periods 2 and 3, and Down Group 1 and Group 17

Category	Elements and Atomic Radii (pm)						
Period 2	Li 152	Be 111	B 88	C 77	N 74	O 66	F 64
Period 3	Na 186	Mg 160	Al 143	Si 117	P 110	S 104	Cl 99
Group 1 (Alkali Metals)	Li 152	Na 186	K 231	Rb 244	Cs 262	–	–
Group 17 (Halogens)	F 64	Cl 99	Br 114	I 133	At 140	–	–

6.2 Ionic Radius and Isoelectronic Species

- **Cations are ALWAYS smaller than their parent atoms ($r_{\text{cation}} < r_{\text{atom}}$):** A cation has fewer electrons while its nuclear charge remains the same, resulting in increased effective nuclear attraction per electron (and often loss of the outermost shell).
  Example: Atomic radius of $Na = 186$ pm, whereas ionic radius of $Na^+ = 95$ pm.

- **Anions are ALWAYS larger than their parent atoms ($r_{\text{anion}} > r_{\text{atom}}$):** Addition of one or more electrons increases electron–electron repulsion and decreases effective nuclear charge, causing the electron cloud to expand.
  Example: Atomic radius of $\mathrm{F} = 64\,\mathrm{pm}$, whereas ionic radius of $\mathrm{F}^- = 136\,\mathrm{pm}$.

> **Isoelectronic Species:** Atoms and ions that contain the **same number of electrons** (or, more broadly, two or more species with the same number of atoms, same number of valence electrons, and same structure).
> For monoatomic isoelectronic ions (e.g., $\mathrm{N}^{3-}, \mathrm{O}^{2-}, \mathrm{F}^-, \mathrm{Ne}, \mathrm{Na}^+, \mathrm{Mg}^{2+}, \mathrm{Al}^{3+}$, all having $10\,\text{electrons}$), ionic radius is inversely proportional to nuclear charge (Z):
> $$\text{Ionic Radius Order: } \mathrm{Al}^{3+} < \mathrm{Mg}^{2+} < \mathrm{Na}^+ < \mathrm{F}^- < \mathrm{O}^{2-} < \mathrm{N}^{3-}$$
> *Rule:* Among isoelectronic species, the cation with the greater positive charge has a smaller radius, while the anion with the greater negative charge has a larger radius.

6.3 Ionization Enthalpy ($\Delta_i H$)

Definition: The energy required to remove an electron from an **isolated gaseous atom (X) in its ground state**. It is expressed in units of $\mathrm{kJ\,mol^{-1}}$.

$$\begin{aligned} \text{First Ionization Enthalpy } (\Delta_i H_1): &\quad \mathrm{X(g)} \rightarrow \mathrm{X^+(g)} + \mathrm{e^-} \\ \text{Second Ionization Enthalpy } (\Delta_i H_2): &\quad \mathrm{X^+(g)} \rightarrow \mathrm{X^{2+}(g)} + \mathrm{e^-} \end{aligned}$$

- **Always Positive:** Energy is always required to remove electrons from an atom, so ionization enthalpies are always positive ($\Delta_i H > 0$).
- **Successive Ionization Enthalpies:** It is more difficult to remove an electron from a positively charged ion than from a neutral atom. Therefore:
  $$\Delta_i H_1 < \Delta_i H_2 < \Delta_i H_3 < \ldots$$
  A massive jump in ionization enthalpy occurs when an electron is removed from a stable, completely filled core noble gas configuration.
- **Maxima and Minima:** In a plot of $\Delta_i H_1$ vs. atomic number Z, **maxima occur at the noble gases** (closed electron shells) and **minima occur at the alkali metals** (single ns^1 electron preceded by a noble gas core, correlating with their high reactivity).

Shielding (Screening) Effect and Periodic Trends in $\Delta_i H$:

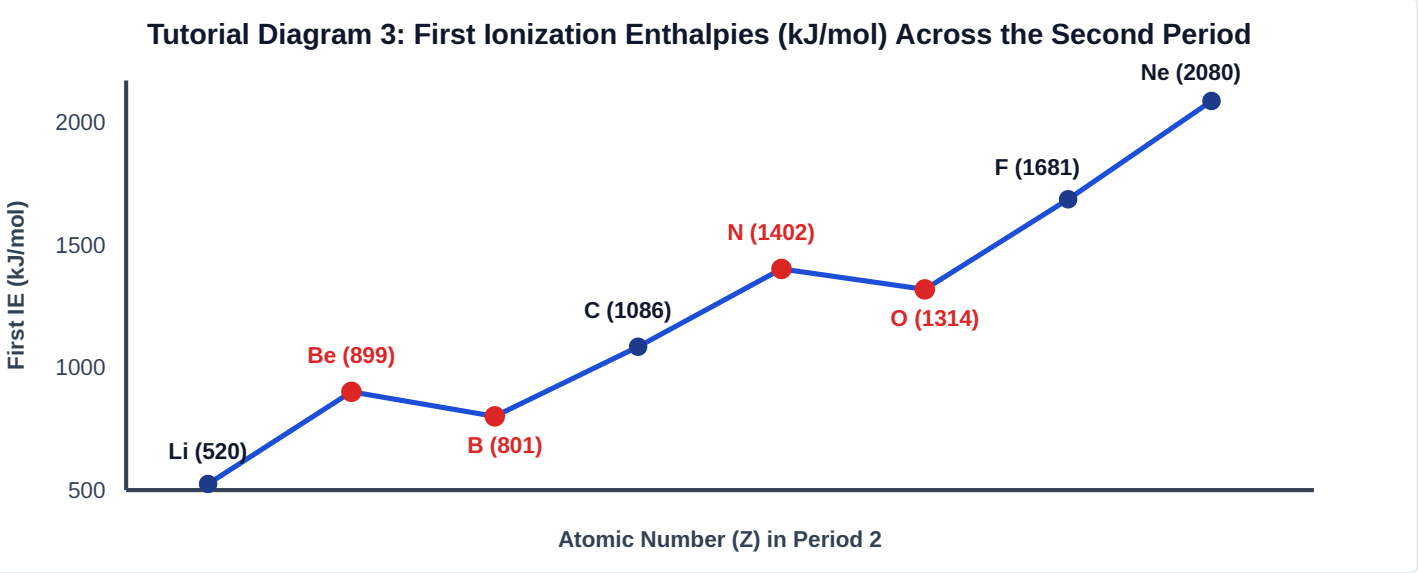
The outer valence electron is shielded (screened) from the nucleus by intervening inner-core electrons, so the **effective nuclear charge (Z_{eff})** experienced by a valence electron is less than the actual nuclear charge (Z).

- **Across a Period: Generally Increases.** Successive electrons are added to the same principal quantum level (n), so shielding does not increase enough to compensate for the increasing nuclear charge.
- **Down a Group: Decreases.** The outermost electron is farther from the nucleus and increased shielding by inner-shell electrons outweighs the increasing nuclear charge.
  Group 1 values ($\mathrm{kJ\,mol^{-1}}$): $\mathrm{Li}\,(520) > \mathrm{Na}\,(496) > \mathrm{K}\,(419) > \mathrm{Rb}\,(403) > \mathrm{Cs}\,(374)$.

Crucial NEET Exceptions in First Ionization Enthalpy ($\Delta_i H_1$):

1. **Period 2 Order ($\mathrm{Be} > \mathrm{B}$ and $\mathrm{N} > \mathrm{O}$):**
   $$\mathrm{Li}\,(520) < \mathrm{B}\,\mathbf{(801)} < \mathrm{Be}\,\mathbf{(899)} < \mathrm{C}\,(1086) < \mathrm{O}\,\mathbf{(1314)} < \mathrm{N}\,\mathbf{(1402)} < \mathrm{F}\,(1681) < \mathrm{Ne}\,(2080)\ \mathrm{kJ\,mol^{-1}}$$
   - **Why $\Delta_i H_1(\mathrm{Be}) > \Delta_i H_1(\mathrm{B})$? (Penetration Effect):** Within the same principal quantum level ($n = 2$), an s-electron penetrates closer to the nucleus than a p-electron ($s > p > d > f$). In $\mathrm{Be}\,(1s^2 2s^2)$, a tightly held $2s$-electron is removed, whereas in $\mathrm{B}\,(1s^2 2s^2 2p^1)$, a more shielded $2p$-electron is removed. (Similarly in Period 3, $\Delta_i H_1(\mathrm{Mg}) > \Delta_i H_1(\mathrm{Al})$).
   - **Why $\Delta_i H_1(\mathrm{N}) > \Delta_i H_1(\mathrm{O})$? (Half-Filled Stability & Pairing Repulsion):** In $\mathrm{N}\,(1s^2 2s^2 2p_x^1 2p_y^1 2p_z^1)$, the three $2p$-electrons reside in different atomic orbitals (Hund's rule, stable half-filled subshell). In $\mathrm{O}\,(1s^2 2s^2 2p_x^2 2p_y^1 2p_z^1)$, two of the four $2p$-electrons occupy the same $2p$-orbital, causing increased **electron–electron repulsion**, making it easier to remove the fourth $2p$-electron. (Similarly in Period 3, $\Delta_i H_1(\mathrm{P}) > \Delta_i H_1(\mathrm{S})$).

2. **Group 13 Ionization Enthalpy Anomaly:**
   Values ($\mathrm{kJ\,mol^{-1}}$): $\mathrm{B}\,(801),\ \mathrm{Al}\,(577),\ \mathrm{Ga}\,(579),\ \mathrm{In}\,(558),\ \mathrm{Tl}\,(589)$.
   Notice the order: $\mathrm{B} > \mathrm{Tl} > \mathrm{Ga} > \mathrm{Al} > \mathrm{In}$.
   Reason: Ga follows the $3d^{10}$ series and Tl follows both the $4f^{14}$ and $5d^{10}$ series. Because inner d- and f-electrons shield the nuclear charge very poorly, the outer valence electrons in Ga and Tl experience a much higher effective nuclear charge (Z_{eff}), causing $\Delta_i H_1(\mathrm{Ga}) > \Delta_i H_1(\mathrm{Al})$ and $\Delta_i H_1(\mathrm{Tl}) > \Delta_i H_1(\mathrm{In})$.



Notice the two characteristic dips in Period 2: at Boron (2p penetration effect vs Be 2s) and at Oxygen (2p electron-pairing repulsion vs half-filled N 2p3).

6.4 Electron Gain Enthalpy ($\Delta_{eg}H$)

Definition: When an electron is added to a **neutral isolated gaseous atom** (X) to convert it into a negative ion, the enthalpy change accompanying the process is defined as the **Electron Gain Enthalpy ($\Delta_{eg}H$)**:

$$\mathrm{X(g)} + \mathrm{e^-} \rightarrow \mathrm{X^-(g)} \quad (\Delta H = \Delta_{eg}H)$$

- **Exothermic or Endothermic:**
  - For most elements, energy is released when an electron is added ($\Delta_{eg}H$ is **negative**). **Group 17 (halogens)** have very high negative $\Delta_{eg}H$ because they attain stable noble gas configurations by picking up one electron.
  - **Noble gases (Group 18)** have **large positive** $\Delta_{eg}H$ because the added electron must enter the next higher principal quantum level, leading to a very unstable electronic configuration.
  - **Second Electron Gain Enthalpy ($\Delta_{eg}H_2$):** Adding a second electron to a negatively charged ion (e.g., $\mathrm{O^-(g)} + \mathrm{e^-} \rightarrow \mathrm{O^{2-}(g)}$) is **always endothermic (positive)** because energy must be supplied to overcome strong electrostatic repulsion between the negatively charged anion and the incoming electron.
- **Relation to Electron Affinity (A_e):** In many texts, the negative of the enthalpy change at absolute zero is called Electron Affinity (A_e). At any temperature T, taking heat capacities into account:

$$\Delta_{eg}H = -A_e - \frac{5}{2}RT$$

Crucial NEET Exception in Electron Gain Enthalpy (Cl vs F, and S vs O):
Generally, $\Delta_{eg}H$ becomes more negative across a period and less negative down a group. However, the **electron gain enthalpy of O or F is LESS negative than that of the succeeding element (S or Cl)**:

- **Group 17 Magnitude of Negative $\Delta_{eg}H$:**
  $\mathrm{Cl}\ (\mathbf{-349}) > \mathrm{F}\ (\mathbf{-328}) > \mathrm{Br}\ (-325) > \mathrm{I}\ (-295) > \mathrm{At}\ (-270)\ \mathrm{kJ\,mol^{-1}}$.
- **Group 16 Magnitude of Negative $\Delta_{eg}H$:**
  $\mathrm{S}\ (-200) > \mathrm{Se}\ (-195) > \mathrm{Te}\ (-190) > \mathrm{Po}\ (-174) > \mathrm{O}\ (\mathbf{-141})\ \mathrm{kJ\,mol^{-1}}$. (Note: Oxygen has the *least negative* $\Delta_{eg}H$ in Group 16!).

Reason: When an electron is added to O or F, it enters the smaller $n = 2$ quantum level ($2p$ subshell) and suffers significant **inter-electronic repulsion** from the other electrons already crowded in this small volume. In S or Cl ($n = 3$ quantum level, $3p$ subshell), the added electron occupies a larger region of space and electron–electron repulsion is much less.

Electron Gain Enthalpies ($\Delta_{eg}H$ in $\mathrm{kJ\,mol^{-1}}$) of Some Main Group Elements

Group 1	$\Delta_{eg}H$	Group 16	$\Delta_{eg}H$	Group 17	$\Delta_{eg}H$	Group 18 (0)	$\Delta_{eg}H$
H	-73	–	–	–	–	He	$+48$
Li	-60	O	-141	F	-328	Ne	$+116$
Na	-53	S	-200	Cl	-349	Ar	$+96$
K	-48	Se	-195	Br	-325	Kr	$+96$
Rb	-47	Te	-190	I	-295	Xe	$+77$
Cs	-46	Po	-174	At	-270	Rn	$+68$

6.5 Electronegativity

Definition: A qualitative measure of the ability of an atom **in a chemical compound** to attract shared electrons to itself.

- Unlike ionization enthalpy and electron gain enthalpy (which are properties of isolated gaseous atoms), electronegativity is **not a directly measurable quantity** and is **not constant** for a given element—it varies depending on the element to which it is bound (and its oxidation state/hybridization).
- **Scales of Electronegativity:** Pauling scale, Mulliken-Jaffe scale, and Allred-Rochow scale. On the widely used **Pauling scale** (1922), **Fluorine (F) is arbitrarily assigned a value of** 4.0 (highest ability to attract electrons).
- **Periodic Trends:** Increases across a period from left to right (as atomic radius decreases) and decreases down a group (as atomic radius increases). Directly proportional to non-metallic character and inversely proportional to metallic character.

Electronegativity Values (on Pauling Scale) Across Periods and Down Groups

Category	Elements and Pauling Electronegativity Values						
Period 2	Li 1.0	Be 1.5	B 2.0	C 2.5	N 3.0	O 3.5	F 4.0
Period 3	Na 0.9	Mg 1.2	Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0
Group 1	Li 1.0	Na 0.9	K 0.8	Rb 0.8	Cs 0.7	–	–
Group 17	F 4.0	Cl 3.0	Br 2.8	I 2.5	At 2.2	–	–

7. Periodic Trends in Chemical Properties and Reactivity

7.1 Periodicity of Valence or Oxidation States

The **valence** of representative elements is usually equal to the **number of electrons in the outermost orbitals (N_v)** and/or equal to $8 - N_v$:

Valence of Representative Elements Across Groups

Group	1	2	13	14	15	16	17	18
Number of valence electrons	1	2	3	4	5	6	7	8
Valence	1	2	3	4	3, 5	2, 6	1, 7	0, 8

Oxidation State: Defined as the charge acquired by an atom in a particular compound on the basis of electronegativity considerations from other atoms in the molecule.

- **Example 1 (OF_2):** Electronegativity order is $F\ (4.0) > O\ (3.5)$. Each $F\ (2s^2 2p^5)$ shares 1 electron with O and is assigned an oxidation state of -1. Oxygen $(2s^2 2p^4)$ shares 2 electrons with the two more electronegative F atoms and exhibits an oxidation state of $+2$.
- **Example 2 (Na_2O):** Electronegativity order is $O\ (3.5) > Na\ (0.9)$. Oxygen accepts two electrons (one from each Na atom) and shows an oxidation state of -2, while $Na\ (3s^1)$ shows $+1$.

Formulas of Hydrides and Oxides Demonstrating Periodic Trends in Valence

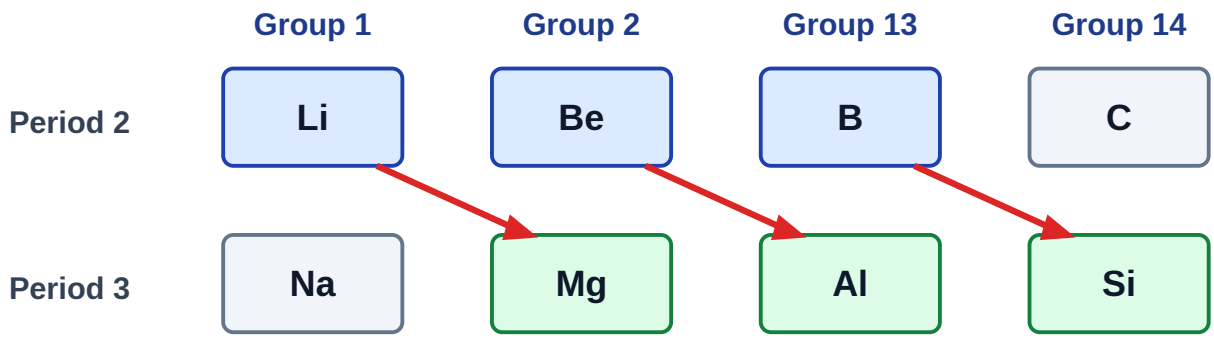
Group	1	2	13	14	15	16	17
Formula of Hydride	LiH NaH KH	CaH_2	B_2H_6 AlH_3	CH_4 SiH_4 GeH_4 SnH_4	NH_3 PH_3 AsH_3	H_2O H_2S H_2Se H_2Te	HF HCl HBr HI
Formula of Oxide	Li_2O Na_2O K_2O	MgO CaO SrO BaO	B_2O_3 Al_2O_3 Ga_2O_3 In_2O_3	CO_2 SiO_2 GeO_2 SnO_2 PbO_2	N_2O_3, N_2O_5 P_4O_6, P_4O_{10} As_2O_3, As_2O_5 Sb_2O_3, Sb_2O_5 Bi_2O_3	SO_3 SeO_3 TeO_3	Cl_2O_7

7.2 Anomalous Properties of Second Period Elements & Diagonal Relationship

The first element of each of Groups 1 (Li), 2 (Be), and 13–17 (B to F) differs significantly in chemical behaviour from the subsequent members of its respective group. For instance, Li and Be form compounds with pronounced covalent character, whereas other members form predominantly ionic compounds.

Diagonal Relationship: The behaviour of Li and Be is more similar to the second element of the following group—namely, Li **resembles** Mg, Be **resembles** Al, and B **resembles** Si. Metallic radii (pm): $Li\ (152)$, $Be\ (111)$, $B\ (88)$ vs. $Na\ (186)$, $Mg\ (160)$, $Al\ (143)$. Ionic radii (pm): $Li^+\ (76)$, $Be^{2+}\ (31)$ vs. $Na^+\ (102)$, $Mg^{2+}\ (72)$.

Tutorial Diagram 4: Diagonal Relationships Between Period 2 and Period 3 Elements



Diagonal similarity in charge/radius ratio (polarizing power) and electronegativity between Period 2 and Period 3 elements.

Four Reasons for the Anomalous Behaviour of Second Period Elements:

1. **Small atomic and ionic size.**
2. **Large charge/radius ratio** (high polarizing power).
3. **High electronegativity.**
4. **Absence of d-orbitals (Maximum Covalency = 4):** The first member of a group has only **four valence orbitals ($2s$ and three $2p$)** available for bonding, whereas the second member ($n = 3$) has **nine valence orbitals ($3s$, three $3p$, and five $3d$)**.
   - Consequently, the **maximum covalency of second-period elements is strictly 4** (e.g., Boron can only form $[BF_4]^-$), whereas third-period elements can expand their valence shell to accommodate more than four pairs of electrons (e.g., Aluminium forms $[AlF_6]^{3-}$, covalency = 6).

5. **Strong p_π–p_π Multiple Bonding:** First members of p-block groups display a much greater ability to form p_π–p_π multiple bonds to themselves (e.g., $\mathrm{C=C}, \mathrm{C\equiv C}, \mathrm{N=N}, \mathrm{N\equiv N}$) and to other second-period elements (e.g., $\mathrm{C=O}, \mathrm{C=N}, \mathrm{C\equiv N}, \mathrm{N=O}$) compared to heavier members of the same group.

7.3 Periodic Trends, Chemical Reactivity, and Nature of Oxides

- **Reactivity Across a Period:** Ionization enthalpy is lowest at the extreme left (alkali metals $\rightarrow$ easy loss of electron $\rightarrow$ strong reducing agents) and electron gain enthalpy is most negative at the extreme right before noble gases (halogens $\rightarrow$ easy gain of electron $\rightarrow$ strong oxidizing agents). Thus, **chemical reactivity is highest at the two extremes of a period and lowest in the centre**.
- **Reactivity Down a Group (Main Group Elements):**
  - **Group 1 (Metals):** Reactivity **increases** down the group ($\mathrm{Li} < \mathrm{Na} < \mathrm{K} < \mathrm{Rb} < \mathrm{Cs}$) because ionization enthalpy decreases down the group, making electron loss progressively easier.
  - **Group 17 (Non-metals):** Reactivity **decreases** down the group ($\mathrm{F} > \mathrm{Cl} > \mathrm{Br} > \mathrm{I}$) because tendency to accept electrons (oxidizing power / electrode potential) decreases down the group.
  - *Note on Transition Metals:* In transition elements, atomic radius changes are much smaller across a period (and even smaller in $4f$ inner-transition metals), ionization enthalpies are intermediate between s- and p-blocks, and down a group a reverse trend in metallic reactivity is observed.
- **Acid–Base Nature of Normal Oxides Across a Period:**
  - **Extreme Left Oxides:** Strongly **Basic** (e.g., $\mathrm{Na_2O}$). Reacts with water to form a strong base:
    $$\mathrm{Na_2O + H_2O \rightarrow 2NaOH}$$
  - **Extreme Right Oxides:** Strongly **Acidic** (e.g., $\mathrm{Cl_2O_7}$). Reacts with water to form a strong acid:
    $$\mathrm{Cl_2O_7 + H_2O \rightarrow 2HClO_4}$$
  - **Centre of Period Oxides:**
    - **Amphoteric Oxides** (behave as acidic with bases and as basic with acids): $\mathrm{Al_2O_3},\ \mathrm{As_2O_3}$ (also $\mathrm{BeO}, \mathrm{ZnO}, \mathrm{Ga_2O_3}$).
    - **Neutral Oxides** (have neither acidic nor basic properties): $\mathrm{CO},\ \mathrm{NO},\ \mathrm{N_2O}$.

8. Comprehensive Worked Examples

Worked Example 1: IUPAC Systematic Nomenclature

Problem: What would be the IUPAC systematic name and symbol for the element with atomic number 120?

Solution:

The digits of atomic number 120 are 1, 2, and 0. Their corresponding IUPAC numerical roots are:
$1 = \mathrm{un\ (u)}, \quad 2 = \mathrm{bi\ (b)}, \quad 0 = \mathrm{nil\ (n)}.$
Putting the roots together in order and adding "-ium" gives the name **unbinilium** and the three-letter symbol Ubn.

Worked Example 2: Quantum Numbers and Period Capacity

Problem: How would you justify the presence of 18 elements in the 5th period and 32 elements in the 6th period of the Periodic Table on the basis of quantum numbers?

Solution:

- **For the 5th Period ($n = 5$):** The available subshells filled in order of increasing energy ($(n + l)$ rule) between the start and end of the 5th period are $5s < 4d < 5p$. The total number of orbitals available is $1\,(5s) + 5\,(4d) + 3\,(5p) = 9\ \mathrm{orbitals}$. Since each orbital accommodates at most 2 electrons, the maximum number of electrons (and hence elements) is $2 \times 9 = 18\ \mathrm{elements}$.
- **For the 6th Period ($n = 6$):** The available subshells filled in order of increasing energy are $6s < 4f < 5d < 6p$. The total number of orbitals available is $1\,(6s) + 7\,(4f) + 5\,(5d) + 3\,(6p) = 16\ \mathrm{orbitals}$. The maximum number of electrons that can be accommodated is $2 \times 16 = 32\ \mathrm{elements}$.

Worked Example 3: Predicting Group and Electronic Configuration of Superheavy Elements

Problem: In which family/group would you place the elements with $Z = 117$ and $Z = 120$, and what is the ground-state electronic configuration in each case?

Solution:

- **For $Z = 117$:** Element $Z = 118$ ($\mathrm{Og/Uuo}$) is the noble gas (Group 18) ending the 7th period. Therefore, $Z = 117$ is one column before Group 18, placing it in the **halogen family (Group 17, Period 7)**. Starting from Radon (Rn, $Z = 86$), its electronic configuration is:
$$[\mathrm{Rn}]\, 5f^{14}\, 6d^{10}\, 7s^2\, 7p^5$$
- **For $Z = 120$:** It has two electrons beyond the 7th-period noble gas ($Z = 118$, $\mathrm{Uuo/Og}$). Thus, it enters the 8th period ($8s^2$) and belongs to **Group 2 (alkaline earth metals)** with electronic configuration:
$$[\mathrm{Uuo}]\, 8s^2 \quad (\text{or } [\mathrm{Og}]\, 8s^2)$$

Worked Example 4: Ordering Metallic Character

Problem: Considering their atomic number and position in the periodic table, arrange the following elements in increasing order of metallic character: $\mathrm{Si}, \mathrm{Be}, \mathrm{Mg}, \mathrm{Na}, \mathrm{P}$.

Solution:

Metallic character increases down a group and decreases across a period from left to right.
In Period 3 (left to right): $\mathrm{Na}\ (\text{Group } 1) > \mathrm{Mg}\ (\text{Group } 2) > \mathrm{Si}\ (\text{Group } 14) > \mathrm{P}\ (\text{Group } 15)$.
In Group 2 (down the group): $\mathrm{Mg}\ (\text{Period } 3) > \mathrm{Be}\ (\text{Period } 2)$, while Be (Group 2 metal) is more metallic than Si (Group 14 metalloid).
Hence, the increasing order of metallic character is:
$$\mathrm{P} < \mathrm{Si} < \mathrm{Be} < \mathrm{Mg} < \mathrm{Na}$$

Worked Example 5: Atomic and Ionic Size Comparison

Problem: Which of the following species will have the largest and the smallest size: $\mathrm{Mg}, \mathrm{Mg}^{2+}, \mathrm{Al}, \mathrm{Al}^{3+}$?

Solution:

Atomic radii decrease across a period ($\mathrm{Mg} > \mathrm{Al}$). Cations are smaller than their parent atoms ($\mathrm{Mg}^{2+} < \mathrm{Mg}$ and $\mathrm{Al}^{3+} < \mathrm{Al}$). Between the isoelectronic cations Mg^{2+} ($Z = 12, 10\ \mathrm{e}^-$) and Al^{3+} ($Z = 13, 10\ \mathrm{e}^-$), the one with the larger positive nuclear charge has a smaller radius ($\mathrm{Al}^{3+} < \mathrm{Mg}^{2+}$).
Overall size order: $\mathrm{Al}^{3+} < \mathrm{Mg}^{2+} < \mathrm{Al} < \mathrm{Mg}$.
Largest species: Mg; **Smallest species:** Al^{3+}.

Worked Example 6: Predicting Ionization Enthalpy of Aluminium

Problem: The first ionization enthalpy ($\Delta_i H$) values of the third-period elements Na, Mg, and Si are respectively $496, 737$, and $786\ \mathrm{kJ\ mol^{-1}}$. Predict whether the first $\Delta_i H$ value for Al will be closer to 575 or $760\ \mathrm{kJ\ mol^{-1}}$. Justify your answer.

Solution:

It will be closer to $575\ \mathrm{kJ\ mol^{-1}}$. Although Al ($Z = 13$, $[\mathrm{Ne}]3s^2 3p^1$) lies between Mg ($Z = 12$, $[\mathrm{Ne}]3s^2$) and Si ($Z = 14$, $[\mathrm{Ne}]3s^2 3p^2$), the value for Al is lower than that of Mg ($737\ \mathrm{kJ\ mol^{-1}}$) because the $3p$-electron removed from Al is effectively shielded from the nucleus by the completely filled $3s^2$ electrons and penetrates less than a $3s$-electron.

Worked Example 7: Comparing Electron Gain Enthalpies

Problem: Which of the following will have the most negative electron gain enthalpy and which the least negative: $\mathrm{P}, \mathrm{S}, \mathrm{Cl}, \mathrm{F}$? Explain your answer.

Solution:

Electron gain enthalpy generally becomes more negative across a period from left to right ($\mathrm{P} < \mathrm{S} < \mathrm{Cl}$ in magnitude of negative value) and less negative down a group. However, adding an electron to the compact $2p$-orbital of F leads to greater inter-electronic repulsion than adding an electron to the larger $3p$-orbital of Cl.
Magnitude of negative $\Delta_{eg}H$: $\mathrm{Cl}\ (-349) > \mathrm{F}\ (-328) > \mathrm{S}\ (-200) > \mathrm{P}$.
Most negative $\Delta_{eg}H$: Cl; **Least negative $\Delta_{eg}H$:** P.

Worked Example 8: Oxidation State vs. Covalency in Coordination Complexes

Problem: Are the oxidation state and covalency of Al in $[\mathrm{AlCl}(\mathrm{H_2O})_5]^{2+}$ the same?

Solution:

No. Since $\mathrm{H_2O}$ is a neutral ligand (0 charge) and $\mathrm{Cl^-}$ has a -1 charge, the oxidation state x of Al satisfies $x + (-1) + 5(0) = +2 \implies x = \mathbf{+3}$. However, Al is bonded to $1\ \mathrm{Cl^-}$ and $5\ \mathrm{H_2O}$ ligands, forming 6 coordinate/covalent bonds, so its **covalency is** 6.

Worked Example 9: Calculating Ionization Enthalpy of Hydrogen from Ground-State Energy

Problem: The energy of an electron in the ground state of the hydrogen atom is $-2.18 \times 10^{-18}\ \mathrm{J}$. Calculate the ionization enthalpy of atomic hydrogen in terms of $\mathrm{J\ mol^{-1}}$ and $\mathrm{kJ\ mol^{-1}}$.

Solution:

The energy required to remove an electron from 1 hydrogen atom in its ground state ($n = 1$) to infinity ($E_\infty = 0$) is:

$$\Delta E_{\text{atom}} = E_\infty - E_1 = 0 - (-2.18 \times 10^{-18}\ \mathrm{J}) = 2.18 \times 10^{-18}\ \mathrm{J\ atom^{-1}}$$

For $1\ \mathrm{mol}$ of hydrogen atoms ($N_A = 6.022 \times 10^{23}\ \mathrm{mol^{-1}}$):

$$\Delta_i H = 2.18 \times 10^{-18}\ \mathrm{J} \times 6.022 \times 10^{23}\ \mathrm{mol^{-1}} = 1.3128 \times 10^{6}\ \mathrm{J\ mol^{-1}} = \mathbf{1312.8}\ \mathrm{kJ\ mol^{-1}}$$

Worked Example 10: Predicting Binary Compound Formulas from Periodic Positions

Problem: Predict the formulas of the stable binary compounds formed by the combination of the following pairs of elements:
(a) Silicon and bromine, (b) Aluminium and sulphur, (c) Lithium and oxygen, (d) Magnesium and nitrogen, (e) Phosphorus and fluorine, (f) Element 71 and fluorine.

Solution:

- **(a) Silicon (Group 14, valence 4) & Bromine (Group 17, valence 1):** $\mathrm{SiBr_4}$
- **(b) Aluminium (Group 13, valence 3) & Sulphur (Group 16, valence 2):** $\mathrm{Al_2S_3}$
- **(c) Lithium (Group 1, valence 1) & Oxygen (Group 16, valence 2):** $\mathrm{Li_2O}$
- **(d) Magnesium (Group 2, valence 2) & Nitrogen (Group 15, valence 3):** $\mathrm{Mg_3N_2}$
- **(e) Phosphorus (Group 15, valence 3 or 5) & Fluorine (Group 17, valence 1):** $\mathrm{PF_3}$ or $\mathrm{PF_5}$
- **(f) Element 71 (Lutetium, Lu, a lanthanoid with common oxidation state/valence $+3$) & Fluorine (valence 1):** $\mathrm{LuF_3}$ (or $\mathrm{EF_3}$)

9. NEET UG Practice Questions with Complete Worked Solutions

Note: All questions below are original exam-aligned practice questions designed strictly for NEET UG preparation based on the chapter syllabus.

Practice Question 1 [Single-Correct MCQ | Electronic Configuration & Periodic Position]

Where would you locate the element with atomic number $Z = 114$ in the modern long form of the Periodic Table, and what is its official IUPAC name?

(A) Period 7, Group 14 (Carbon family) — Flerovium (Fl)

(B) Period 7, Group 16 (Chalcogen family) — Livermorium (Lv)

(C) Period 6, Group 14 (Carbon family) — Copernicium (Cn)

(D) Period 7, Group 15 (Nitrogen family) — Moscovium (Mc)

Correct Answer: (A)

Worked Solution: The 7th period ends at the noble gas with $Z = 118$ (Group 18). Counting backwards from Group 18 ($Z = 118$), element $Z = 114$ lies in **Period 7** and **Group** $18 - (118 - 114) = 14$ (the Carbon family, p-block), with outer configuration $[\mathrm{Rn}]\, 5f^{14}\, 6d^{10}\, 7s^2\, 7p^2$. Its systematic name is Ununquadium (Uuq) and its official IUPAC name is **Flerovium** (Fl).

Practice Question 2 [Single-Correct MCQ | Structure of Modern Periodic Table]

Which of the following statements related to the modern periodic table is **incorrect**?

(A) The p-block has 6 columns because a maximum of 6 electrons can occupy all the orbitals in a p-subshell.

(B) The d-block has 8 columns because a maximum of 8 electrons can occupy all the orbitals in a d-subshell.

(C) Each block contains a number of columns equal to the maximum number of electrons that can occupy that subshell.

(D) The block indicates the value of the azimuthal quantum number (l) for the last subshell that received electrons in building up the electronic configuration.

Correct Answer: (B)

Worked Solution: A d-subshell $(l = 2)$ has $2l + 1 = 5$ orbitals, which can accommodate a maximum of $2 \times 5 = \mathbf{10\ electrons}$. Therefore, the d-block has **10 columns** (Groups 3 to 12), not 8 columns. Hence, statement (B) is incorrect.

Practice Question 3 [Single-Correct MCQ | Isoelectronic Species & Ionic Radii]

Consider the following isoelectronic species: $\mathrm{N}^{3-}, \mathrm{O}^{2-}, \mathrm{F}^{-}, \mathrm{Na}^{+}, \mathrm{Mg}^{2+}$ and Al^{3+}. Which of the following represents the correct order of their **increasing ionic radii**?

(A) $\mathrm{N}^{3-} < \mathrm{O}^{2-} < \mathrm{F}^{-} < \mathrm{Na}^{+} < \mathrm{Mg}^{2+} < \mathrm{Al}^{3+}$

(B) $\mathrm{Al}^{3+} < \mathrm{Mg}^{2+} < \mathrm{Na}^{+} < \mathrm{F}^{-} < \mathrm{O}^{2-} < \mathrm{N}^{3-}$

(C) $\mathrm{Na}^{+} < \mathrm{Mg}^{2+} < \mathrm{Al}^{3+} < \mathrm{F}^{-} < \mathrm{O}^{2-} < \mathrm{N}^{3-}$

(D) $\mathrm{Al}^{3+} < \mathrm{Mg}^{2+} < \mathrm{Na}^{+} < \mathrm{N}^{3-} < \mathrm{O}^{2-} < \mathrm{F}^{-}$

Correct Answer: (B)

Worked Solution: All six species are isoelectronic, each containing 10 electrons $(1s^2 2s^2 2p^6)$. Their nuclear charges (Z) are: $\mathrm{N}^{3-}\ (7), \mathrm{O}^{2-}\ (8), \mathrm{F}^{-}\ (9), \mathrm{Na}^{+}\ (11), \mathrm{Mg}^{2+}\ (12), \mathrm{Al}^{3+}\ (13)$. As nuclear charge increases for the same number of electrons, attraction of the electrons to the nucleus increases and ionic radius decreases. Thus, the increasing order of ionic radii is $\mathrm{Al}^{3+} < \mathrm{Mg}^{2+} < \mathrm{Na}^{+} < \mathrm{F}^{-} < \mathrm{O}^{2-} < \mathrm{N}^{3-}$.

Practice Question 4 [Single-Correct MCQ | Ionization Enthalpy Principles]

Which one of the following statements is **incorrect** in relation to ionization enthalpy?

(A) Ionization enthalpy increases for each successive electron removed.

(B) The greatest increase in ionization enthalpy is experienced on removal of an electron from a core noble gas configuration.

(C) The end of valence electrons is marked by a big jump in ionization enthalpy.

(D) Removal of an electron from orbitals bearing a lower n value is easier than from an orbital having a higher n value.

Correct Answer: (D)

Worked Solution: Orbitals with a lower principal quantum number (n) are closer to the nucleus and experience much stronger nuclear attraction and less shielding than orbitals with a higher n value. Therefore, removal of an electron from orbitals bearing a lower n value is **more difficult** (requires more energy) than from orbitals having a higher n value. Statement (D) is incorrect.

Practice Question 5 [Data-Analysis Table Question | $\Delta_i H_1$, $\Delta_i H_2$, and $\Delta_{eg} H$]

The first ($\Delta_i H_1$) and second ($\Delta_i H_2$) ionization enthalpies (in $\mathrm{kJ\,mol^{-1}}$) and the electron gain enthalpy ($\Delta_{eg} H$ in $\mathrm{kJ\,mol^{-1}}$) of six elements I to VI are given below:

Element	$\Delta_i H_1$ ($\mathrm{kJ\,mol^{-1}}$)	$\Delta_i H_2$ ($\mathrm{kJ\,mol^{-1}}$)	$\Delta_{eg} H$ ($\mathrm{kJ\,mol^{-1}}$)
I	520	7300	-60
II	419	3051	-48
III	1681	3374	-328
IV	1008	1846	-295
V	2372	5251	$+48$
VI	738	1451	-40

Match the following descriptions with the correct elements from the table:

- (i) The least reactive element
- (ii) The most reactive metal
- (iii) The most reactive non-metal
- (iv) The metal that forms a stable binary halide of the formula $\mathrm{MX_2}$ ($\mathrm{X = halogen}$)
- (v) The metal that forms a predominantly stable covalent halide of the formula MX ($\mathrm{X = halogen}$)

(A) (i)–V, (ii)–II, (iii)–III, (iv)–VI, (v)–I

(B) (i)–V, (ii)–I, (iii)–III, (iv)–VI, (v)–II

(C) (i)–IV, (ii)–II, (iii)–III, (iv)–I, (v)–VI

(D) (i)–V, (ii)–II, (iii)–IV, (iv)–VI, (v)–I

Correct Answer: (A)

Worked Solution:

- **(i) Least reactive element = V:** Has the highest $\Delta_i H_1$ ($2372\,\mathrm{kJ\,mol^{-1}}$) and a positive $\Delta_{eg} H$ ($+48\,\mathrm{kJ\,mol^{-1}}$), characteristic of a noble gas (Helium).
- **(ii) Most reactive metal = II:** Has the lowest $\Delta_i H_1$ ($419\,\mathrm{kJ\,mol^{-1}}$) and a huge jump to $\Delta_i H_2$ ($3051\,\mathrm{kJ\,mol^{-1}}$), identifying it as a heavier alkali metal (Potassium, K).
- **(iii) Most reactive non-metal = III:** Has a very high $\Delta_i H_1$ ($1681\,\mathrm{kJ\,mol^{-1}}$) and the most negative $\Delta_{eg} H$ ($-328\,\mathrm{kJ\,mol^{-1}}$), identifying it as Fluorine (F). (Note: IV is Iodine, the least reactive non-metal among the halogens shown).
- **(iv) Metal forming stable $\mathrm{MX_2}$ = VI:** Both $\Delta_i H_1$ (738) and $\Delta_i H_2$ (1451) are relatively low with no massive jump between $\Delta_i H_1$ and $\Delta_i H_2$, identifying it as an alkaline earth metal (Magnesium, Mg).
- **(v) Metal forming predominantly covalent MX = I:** Has low $\Delta_i H_1$ (520) and a massive jump to $\Delta_i H_2$ (7300), identifying it as Lithium (Li), which forms predominantly covalent halides (LiX) due to its small size and high polarizing power.

Practice Question 6 [Assertion-Reason | Second Ionization Enthalpy of Na vs. Mg]

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): The first ionization enthalpy of sodium (Na) is lower than that of magnesium (Mg), but the second ionization enthalpy of sodium is much higher than that of magnesium.

Reason (R): In Na^{+}, the second electron has to be removed from a stable, completely filled core noble gas configuration $(1s^2 2s^2 2p^6)$, whereas in Mg^{+}, the second electron is removed from the $3s^1$ valence orbital.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Correct Answer: (A)

Worked Solution: For neutral atoms, $\mathrm{Na}\,(1s^2 2s^2 2p^6 3s^1)$ has a lower nuclear charge and larger atomic radius than $\mathrm{Mg}\,(1s^2 2s^2 2p^6 3s^2)$, and removing one $3s$ electron from Na yields a stable neon core; hence $\Delta_i H_1(\mathrm{Na}) < \Delta_i H_1(\mathrm{Mg})$. However, once Na^{+} $(1s^2 2s^2 2p^6)$ is formed, removing a second electron disrupts the stable noble gas $(n = 2)$ core, whereas removing a second electron from Mg^{+} $(1s^2 2s^2 2p^6 3s^1)$ removes the remaining $3s^1$ electron to achieve the stable neon core. Thus both (A) and (R) are true and (R) correctly explains (A).

Practice Question 7 [Assertion-Reason | Second Electron Gain Enthalpy of Oxygen]

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): The first electron gain enthalpy of oxygen is negative $(-141\ \mathrm{kJ\ mol^{-1}})$, whereas the second electron gain enthalpy of oxygen to form $\mathrm{O}^{2-}(\mathrm{g})$ is positive.

Reason (R): When a second electron is added to the mono-negative $\mathrm{O}^{-}(\mathrm{g})$ ion, energy must be supplied to overcome the strong electrostatic repulsion between the negatively charged O^{-} ion and the incoming electron.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Correct Answer: (A)

Worked Solution: Adding the first electron to a neutral oxygen atom $(\mathrm{O(g)} + \mathrm{e}^{-} \rightarrow \mathrm{O}^{-}(\mathrm{g}))$ releases energy ($\Delta_{eg} H_1 = -141\ \mathrm{kJ\ mol^{-1}}$). However, adding a second electron to an already negatively charged $\mathrm{O}^{-}(\mathrm{g})$ ion ($\mathrm{O}^{-}(\mathrm{g}) + \mathrm{e}^{-} \rightarrow \mathrm{O}^{2-}(\mathrm{g})$) encounters strong electrostatic repulsion, requiring an input of energy ($\Delta_{eg} H_2 > 0$, endothermic). Both (A) and (R) are true and (R) is the exact explanation of (A).

Practice Question 8 [Matching Type | Outer Electronic Configuration & Periodic Position]

Match the outer electronic configurations given in **List I** with their correct periodic positions in **List II**:

List I (Outer Electronic Configuration)	List II (Position in Periodic Table)
(P) ns^2np^4 for $n = 3$	(1) Period 6, Group 3 (f-block, Lanthanoid: Gadolinium, $Z = 64$)
(Q) $(n-1)d^2ns^2$ for $n = 4$	(2) Period 3, Group 16 (p-block: Sulphur, $Z = 16$)
(R) $(n-2)f^7(n-1)d^1ns^2$ for $n = 6$	(3) Period 5, Group 10 (d-block: Palladium, $Z = 46$)
(S) $(n-1)d^{10}ns^0$ for $n = 5$	(4) Period 4, Group 4 (d-block: Titanium, $Z = 22$)

(A) P–2, Q–4, R–1, S–3

(B) P–2, Q–3, R–1, S–4

(C) P–4, Q–2, R–1, S–3

(D) P–2, Q–4, R–3, S–1

Correct Answer: (A)

Worked Solution:

- **(P)** $3s^2 3p^4$: Highest $n = 3 \implies$ Period 3; p-block group $= 10 + (2 + 4) =$ Group 16 (Sulphur, $Z = 16$) $\rightarrow$ **(2)**.
- **(Q)** $3d^2 4s^2$: Highest $n = 4 \implies$ Period 4; d-block group $= 2 + 2 =$ Group 4 (Titanium, $Z = 22$) $\rightarrow$ **(4)**.
- **(R)** $4f^7 5d^1 6s^2$: Highest $n = 6 \implies$ Period 6; f-block lanthanoid series (all placed in Group 3; Gadolinium, $Z = 54 + 2 + 1 + 7 = 64$) $\rightarrow$ **(1)**.
- **(S)** $4d^{10}5s^0$: Exceptional configuration of Palladium (Pd, $Z = 46$) in Period 5, Group 10 $\rightarrow$ **(3)**.

Practice Question 9 [Multiple-Statement Evaluation | Group 13 Ionization Enthalpy Trend]

The first ionization enthalpy ($\Delta_i H_1$) values (in $\mathrm{kJ\,mol^{-1}}$) of Group 13 elements are:
$\mathrm{B}\,(801), \mathrm{Al}\,(577), \mathrm{Ga}\,(579), \mathrm{In}\,(558), \mathrm{Tl}\,(589)$. Consider the following statements regarding this trend:

1. The correct decreasing order of first ionization enthalpy in Group 13 is $\mathrm{B} > \mathrm{Tl} > \mathrm{Ga} > \mathrm{Al} > \mathrm{In}$.
2. The sharp decrease from B to Al is due to the increase in atomic size and increased shielding by inner electrons.
3. Ga has a slightly higher $\Delta_i H_1$ than Al because of the poor shielding of the outer electrons by the intervening $3d^{10}$ electrons in Ga.
4. Tl has a higher $\Delta_i H_1$ than In and Al because of the poor shielding provided by the inner $4f^{14}$ and $5d^{10}$ electrons in Tl.

Which of the statements given above are **correct**?

(A) 1 and 2 only

(B) 2, 3 and 4 only

(C) 1, 3 and 4 only

(D) 1, 2, 3 and 4

Correct Answer: (D)

Worked Solution: Comparing the numerical values $801\,(\mathrm{B}) > 589\,(\mathrm{Tl}) > 579\,(\mathrm{Ga}) > 577\,(\mathrm{Al}) > 558\,(\mathrm{In})$, statement 1 is completely accurate. Statement 2 explains the normal decrease from Period 2 (B) to Period 3 (Al). Statements 3 and 4 explain the anomalies at Ga (due to poor shielding by $3d^{10}$ electrons) and Tl (due to poor shielding by $4f^{14}$ and $5d^{10}$ electrons, increasing Z_{eff}). All four statements (1, 2, 3, and 4) are correct.

Practice Question 10 [Single-Correct MCQ | Metallic, Non-Metallic & Oxidizing Orders]

Which of the following sets of periodic property orders is **completely correct**?

1. **Metallic character:** $\mathrm{K} > \mathrm{Mg} > \mathrm{Al} > \mathrm{B}$
2. **Non-metallic character:** $\mathrm{F} > \mathrm{N} > \mathrm{C} > \mathrm{B} > \mathrm{Si}$
3. **Chemical reactivity in terms of oxidizing property:** $\mathrm{F} > \mathrm{Cl} > \mathrm{O} > \mathrm{N}$

(A) 1 and 2 only

(B) 1 and 3 only

(C) 2 and 3 only

(D) 1, 2 and 3

Correct Answer: (D)

Worked Solution:

- **Order 1:** Metallic character increases down a group and decreases left to right across a period. Thus K (Period 4, Group 1) $>$ Mg (Period 3, Group 2) $>$ Al (Period 3, Group 13) $>$ B (Period 2, Group 13). Correct!
- **Order 2:** Non-metallic character increases left to right across Period 2 ($\mathrm{F} > \mathrm{N} > \mathrm{C} > \mathrm{B}$) and decreases down Group 14/13 ($\mathrm{B} > \mathrm{Si}$). Thus $\mathrm{F} > \mathrm{N} > \mathrm{C} > \mathrm{B} > \mathrm{Si}$. Correct!
- **Order 3:** Oxidizing power depends on standard reduction potential and tendency to gain electrons in chemical reactions. Although Oxygen has higher Pauling electronegativity (3.5) than Chlorine (3.0), Chlorine has a much more negative electron gain enthalpy ($-349\,\mathrm{kJ\,mol^{-1}}$) and higher oxidizing power in aqueous/chemical reactions than $\mathrm{O_2}$, giving the oxidizing reactivity order $\mathrm{F} > \mathrm{Cl} > \mathrm{O} > \mathrm{N}$. Correct!

Practice Question 11 [Multiple-Statement Evaluation | Fundamental Concepts of Periodicity]

Read the following four statements carefully:

1. The first ionization enthalpies of two isotopes of the same element are expected to be the same because they have the same atomic number, nuclear charge, and electronic configuration.
2. Nuclear mass does not affect the valence shell electrons or the chemical properties of an element.
3. The electronegativity of nitrogen (N) on the Pauling scale is strictly constant at 3.0 in all nitrogen compounds regardless of oxidation state or hybridization.
4. In defining ionization enthalpy and electron gain enthalpy, the terms "isolated gaseous atom" and "ground state" are specified so that interatomic interactions are absent and the atom is in its most stable state for valid comparison.

Which of the above statements are **correct**?

(A) 1, 2 and 4 only

(B) 1, 2 and 3 only

(C) 2, 3 and 4 only

(D) 1, 2, 3 and 4

Correct Answer: (A)

Worked Solution: Statements 1, 2, and 4 are true. Statement 3 is **false**: the electronegativity of any given element is *not constant*; it varies depending on the element to which it is bound, its oxidation state, and its state of hybridization (e.g., N has different effective electronegativities in $\mathrm{NH_3}$, $\mathrm{NO_2}$, and HCN).

Practice Question 12 [Single-Correct MCQ | Chemical Nature of Oxides & Group 17 Properties]

Identify the **correct** pair of statements from the following:

1. Among $\mathrm{Na_2O}$, $\mathrm{Al_2O_3}$, $\mathrm{Cl_2O_7}$, CO, and $\mathrm{As_2O_3}$, the amphoteric oxides are $\mathrm{Al_2O_3}$ and $\mathrm{As_2O_3}$, while CO is a neutral oxide.
2. Group 17 (halogen family) contains a non-metal gas ($\mathrm{F_2}$, $\mathrm{Cl_2}$), a liquid ($\mathrm{Br_2}$), a solid ($\mathrm{I_2}$), and metallic/metalloid character in its heaviest members at room temperature.
3. Boron can form the hexacoordinate complex ion $[\mathrm{BF_6}]^{3-}$ just like aluminium forms $[\mathrm{AlF_6}]^{3-}$.

(A) 1 and 2 only

(B) 2 and 3 only

(C) 1 and 3 only

(D) 1, 2 and 3

Correct Answer: (A)

Worked Solution: Statement 1 is true ($\mathrm{Al_2O_3}$ and $\mathrm{As_2O_3}$ are amphoteric; $\mathrm{CO}, \mathrm{NO}, \mathrm{N_2O}$ are neutral; $\mathrm{Na_2O}$ is basic; $\mathrm{Cl_2O_7}$ is acidic). Statement 2 is true (Group 17 spans gases, liquid bromine, solid iodine, and heavier metalloid/metallic elements). Statement 3 is **false** because Boron is in Period 2 ($n = 2$) and has only four valence orbitals ($2s$ and three $2p$), restricting its maximum covalency to 4 ($[\mathrm{BF_4}]^-$ only, never $[\mathrm{BF_6}]^{3-}$).

10. Common NEET Pitfalls and Final Revision Checklist

Top 7 Exam Pitfalls to Avoid in NEET UG

1. **Confusing F and Cl in Electron Gain Enthalpy vs. Electronegativity vs. Oxidizing Power:**
   - **Electronegativity:** $\mathrm{F}\ (4.0) > \mathrm{Cl}\ (3.0)$
   - **Oxidizing Power:** $\mathrm{F_2} > \mathrm{Cl_2}$
   - **Magnitude of Negative Electron Gain Enthalpy ($\Delta_{eg}H$):** $\mathrm{Cl}\ (\mathbf{-349\,kJ\,mol^{-1}}) > \mathrm{F}\ (\mathbf{-328\,kJ\,mol^{-1}})$ (due to $2p$–$2p$ inter-electronic repulsion in small F).
2. **Oxygen's Position in Group 16 Electron Gain Enthalpy:** Oxygen does not merely have a less negative $\Delta_{eg}H$ than Sulphur; it has the **least negative $\Delta_{eg}H$ in the entire Group 16**: $\mathrm{S}\ (-200) > \mathrm{Se}\ (-195) > \mathrm{Te}\ (-190) > \mathrm{Po}\ (-174) > \mathrm{O}\ \mathbf{(-141)}\,\mathrm{kJ\,mol^{-1}}$.
3. **Adjacent Ionization Enthalpy Reversals:** Never write a strictly monotonic increase across Period 2 or Period 3. Always swap Group 2 with Group 13 ($\mathrm{Be} > \mathrm{B}$ and $\mathrm{Mg} > \mathrm{Al}$) and Group 15 with Group 16 ($\mathrm{N} > \mathrm{O}$ and $\mathrm{P} > \mathrm{S}$) for Δ_iH_1.
4. **Group 13 Δ_iH_1 Zig-Zag Order:** Memorize the exact order $\mathrm{B} > \mathrm{Tl} > \mathrm{Ga} > \mathrm{Al} > \mathrm{In}$ (values: $801 > 589 > 579 > 577 > 558\,\mathrm{kJ\,mol^{-1}}$).
5. **Palladium's Ground-State Configuration:** Unlike other d-block elements with ns^{1-2}, Palladium (Pd, $Z = 46$) has **zero** outer s-electrons: $[\mathrm{Kr}]\,4d^{10}\,5s^0$.
6. $\mathrm{Zn}, \mathrm{Cd}, \mathrm{Hg}$ **Classification:** They are d-block elements (Group 12), **not** transition elements, because their d-orbitals are completely filled ($(n-1)d^{10}ns^2$) in the ground state as well as in their common oxidation states.
7. **Covalency vs. Oxidation State of Period 2 vs. Period 3 Elements:** Period 2 elements (Li to F) can never exceed a covalency of 4 due to the absence of $2d$ orbitals.

Concise Final Recap

- **Mendeleev's Periodic Law:** Properties of elements are a periodic function of their *atomic weights*. (Predicted Eka-aluminium = Gallium, Eka-silicon = Germanium).
- **Modern Periodic Law (Moseley, 1913):** Physical and chemical properties of elements are periodic functions of their *atomic numbers (Z)* ($\sqrt{\nu} \propto Z$).
- **Long Form Periodic Table:** 7 horizontal periods (indicating highest principal quantum number n; sizes: $2, 8, 8, 18, 18, 32, 32$) and 18 vertical groups.

- **Four Blocks:**
  - s-block (Groups 1–2): ns^{1-2}
  - p-block (Groups 13–18): ns^2np^{1-6} (s and p together = Representative / Main Group elements)
  - d-block (Groups 3–12): $(n-1)d^{1-10}ns^{0-2}$ (Transition elements, except $\mathrm{Zn}, \mathrm{Cd}, \mathrm{Hg}$)
  - f-block (Bottom two rows, Group 3): $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$ (Inner-transition elements: $4f$ Lanthanoids $Z = 58$–71; $5f$ Actinoids $Z = 90$–103).
- **Across a Period (Left $\rightarrow$ Right):** Atomic/ionic radius decreases; $\Delta_i H$, magnitude of negative $\Delta_{eg} H$, electronegativity, non-metallic character, and oxide acidity increase.
- **Down a Group (Top $\rightarrow$ Bottom):** Atomic/ionic radius, metallic character, and oxide basicity increase; $\Delta_i H$, magnitude of negative $\Delta_{eg} H$, and electronegativity decrease.